

Lab 1: Knowledge Is Power

Brendyn Macaluso

CS 411W

Professor T. Kennedy

1 September 2026

Draft 0

Table of Contents

1. Introduction	4
1.1 Societal Problem	4
1.2 Needed Solution Characteristics	4
1.3 Product Introduction	4
2. KIP (Knowledge Is Power) Product Description	5
2.1 Key Product Features and Capabilities	5
2.1.1 Solution Overview	5
2.1.2 Goals and Objectives	5
2.1.3 Interactive Gamification	5
2.1.3.1 “Fact or Fiction?” Game	6
2.1.3.2 “Choose the Outcome” Game	6
2.1.4 Progress Tracking and Analytics	6
2.2 Major Components (Hardware/Software)	7
2.2.1 Presentation Layer	7
2.2.1.1 Tech Stack	7
2.2.1.2 Account Portal	7
2.2.1.3 Module Browser	7
2.2.1.4 Game Browser	7
2.2.1.5 Scenario Player	7
2.2.2 Application Layer	7
2.2.2.1 Tech Stack	8
2.2.2.2 Authentication System	8

Lab 1 – Knowledge Is Power	3
2.2.2.3 Scenario Engine	8
2.2.2.4 Feedback System	8
2.2.2.5 Curation System	8
2.2.2.6 Source Reference System	8
2.2.3 Data Layer	9
2.2.3.1 Tech Stack	9
2.2.3.2 Databases	9
2.2.3.3 Functions	9
2.2.4 Hardware Layer	9
2.2.4.1 Server-Side Components	9
2.2.4.2 Client-Side Components	10
2.2.4.3 Functions	10
3. Identification of Case Study	10
3.1 Primary Target Audience	10
3.2 Secondary Stakeholders	10
3.3 Future Users	10
4. Glossary	11
5. References	12

List of Figures

No figures right now

List of Tables

No tables right now

1. Introduction

This document introduces Knowledge Is Power (KIP), a digital legal literacy platform for young adults. It describes the societal problem KIP addresses, the characteristics a solution to that problem needs, and a brief introduction to the product.

1.1 Societal Problem

Young people ages 16–19 increasingly encounter laws involving driving, digital conduct, work, school, ownership, contracts, and safety without fully understanding their inner workings and the consequences that may ensue. As they gain more independence into adulthood, this lack of legal awareness can lead to poor decisions and serious consequences. Despite this, law-related education remains largely absent from K-12 curricula, leaving young adults underprepared for real-world legal situations. Teen drivers face higher safety risks that are often linked to poor judgement and decision-making (Centers for Disease Control and Prevention, 2026; Legal Services Corporation, 2017; National Highway Traffic Safety Administration, 2024; Touro Law Jacob D. Fuchsberg Law Center, 2017).

1.2 Needed Solution Characteristics

The platform should teach legal concepts through interactive activities that ask users to apply laws to realistic situations rather than simply memorize definitions. To remain accessible, it should use clear and consistent language, concise explanations, mobile-friendly web design, and low-stakes practice games that students can complete independently. It should also explain the connection between each lesson and its official legal sources while introducing legal vocabulary in context so that users can recognize practical terms used across different legal topics.

1.3 Product Introduction

Knowledge Is Power (KIP) is a digital legal literacy platform designed to teach essential, everyday laws to young adults. It serves as a high-quality supplemental educational aid for parents, teachers, and school administrators. KIP's goal is to help young adults recognize legally significant situations and understand the potential consequences of their choices, in order to prepare them to make more informed decisions. The prototype will demonstrate these goals through a web-based service that supports user accounts, learning modules, interactive scenarios, progress tracking, and recommendation features based on user performance.

2. KIP (Knowledge Is Power) Product Description

KIP is structured as a web-based educational service built around learning modules, interactive games, user progress data, and source-backed legal content. The system presents users with short misconception-based quizzes and longer branching scenarios, then uses their performance to recommend future practice. Its software components support account access, scenario delivery, feedback generation, content curation, source referencing, and persistent progress tracking. The goal of these features is to assess users' current legal understanding, strengthen their recognition of legal situations, and reinforce informed decision-making through repeated low-stakes practice.

2.1 Key Product Features and Capabilities

This section outlines KIP's key features and capabilities, including its overall solution, its goals and objectives, its interactive games, and its progress tracking and analytics.

2.1.1 Solution Overview

KIP uses focused games, real-world examples, curated media, and official resources to educate young adults on practical legal boundaries.

2.1.2 Goals and Objectives

KIP aims to assess, improve, and reinforce legal knowledge among tech-friendly youth through repeatable, low-stakes interactive activities. Its user experience (UX) is designed to increase recognition of legal situations, improve decision-making skills, and strengthen long-term retention of legal concepts.

2.1.3 Interactive Gamification

KIP uses focused, low-stakes games to improve users' confidence in making legal decisions.

2.1.3.1 “Fact or Fiction?” Game

This short-form game presents users with common legal beliefs and misconceptions, offering “true”, “false”, or “context-dependent” as answers. This approach will help users challenge assumptions, spot legal nuance, and build foundational knowledge to prepare them for longer and more elaborate scenarios.

2.1.3.2 “Choose the Outcome” Game

This multiple-choice, long-form game encourages users to interact with people, objects, and documents to learn scenario details presented to them before making a decision. Every choice will demonstrate its consequences to illuminate its legal significance, which encourages users to slow down, evaluate context, and understand what information an outcome is dependent on.

2.1.4 Progress Tracking and Analytics

KIP tracks topical strengths, weaknesses, and completion during use. This information allows KIP to recommend relevant games and modules while helping each game session offer questions, scenarios, and legal concepts that meaningfully test the user.

2.2 Major Components (Hardware/Software)

KIP's major components are organized into four layers: the presentation layer, the application layer, the data layer, and the hardware layer. Each layer is described below.

2.2.1 Presentation Layer

The presentation layer is the part of KIP that users see and interact with in their web browser. Its technology and components are described below.

2.2.1.1 Tech Stack

The presentation layer is implemented using JavaScript, Bootstrap, and HTML5.

2.2.1.2 Account Portal

This system allows established users to log in, and newcomers to create an account or play games without an account to get a taste of KIP, albeit with limited features.

2.2.1.3 Module Browser

This browser displays available legal learning modules such as traffic law, online conduct, school and work, and personal safety. It allows users to select a topic and view their progress status for that topic's module.

2.2.1.4 Game Browser

This browser displays the available games and prioritizes recommendations based on the user's progress, performance history, and demonstrated strengths or weaknesses in the legal subtopics covered by each game.

2.2.1.5 Scenario Player

This system is the sole interface users interact with while playing KIP's games. It presents interactive questions, branching scenarios, feedback, and source references during gameplay.

2.2.2 Application Layer

The application layer contains the logic that powers KIP's features, including account access, scenario delivery, feedback generation, content curation, and source referencing. Its technology and components are described below.

2.2.2.1 Tech Stack

The application layer is implemented using Java and JDBC.

2.2.2.2 Authentication System

This system manages user login, account creation, guest access, and session validation. It determines whether a user is using KIP as a registered account or as a limited guest, and it controls which features are available based on that status.

2.2.2.3 Scenario Engine

This system loads and manages the interactive content used during KIP's games. For a play session, it assembles a "match" of play, with multiple "rounds." Each round establishes unique choices, scenario branches, consequences, and game states that appear in sequence. This is primarily informed by the selected game and the user's previous actions.

2.2.2.4 Feedback System

This system provides explanations after user decisions. It identifies whether an answer is correct, incorrect, or context-dependent, explaining the legal reasoning behind the result in order to help the user understand why a specific choice was safe, risky, or incomplete.

2.2.2.5 Curation System

This system organizes and recommends learning content based on the user's progress and performance history. It determines what games, modules, and scenarios to prioritize, depending

on where the user needs more practice, while also allowing users to freely browse and ultimately select any available content.

2.2.2.6 Source Reference System

This system connects game content and feedback to official legal sources, educational resources, and relevant vocabulary. It allows users to see where KIP's information comes from and, when desired, review the original legal language and documents behind a lesson. It also supports content maintenance by tracking changes to legislation, updating references when needed, and flagging KIP content that may be outdated so administrators can review it, revise it, or disable access to it.

2.2.3 Data Layer

The data layer is responsible for the persistent storage of KIP's information. Its technology, databases, and functions are described below.

2.2.3.1 Tech Stack

The data layer is implemented using SQLite.

2.2.3.2 Databases

KIP utilizes a User Database, a Progress Database, a Module Database, a Game Database, and a Legal Content Database.

2.2.3.3 Functions

The data layer stores user account information, stores user progress records, stores learning modules and associated content, stores game scenario questions and gameplay information, and stores legal and educational resources.

2.2.4 Hardware Layer

The hardware layer provides the physical and virtual infrastructure needed to host and access KIP. It includes both server-side and client-side components, which are described below.

2.2.4.1 Server-Side Components

A virtual private server hosts the Java application, manages server-side processing, and provides access to the SQLite database file and application resources.

2.2.4.2 Client-Side Components

Consumer devices with a modern web browser and an internet connection are required to access KIP.

2.2.4.3 Functions

The hardware layer runs the Java application code, stores persistent data, processes user requests, and delivers the web-based interface to users.

3. Identification of Case Study

This section identifies the audiences KIP is designed to serve, including its primary target audience, its secondary stakeholders, and its potential future users.

3.1 Primary Target Audience

KIP's primary target audience is young people aged 16–19 who are preparing to enter adulthood and navigate high-risk legal areas, such as driving, work, and substance use. This group is well suited for KIP because its members are old enough for legal consequences to become personally significant, but still young enough to be especially receptive to education, habit formation, and the correction of misconceptions. By focusing on users at this transitional stage, KIP can help build foundational legal awareness and encourage informed decision-making before risky habits or misunderstandings become more difficult to change.

3.2 Secondary Stakeholders

Parents, teachers, and high school administrators may use KIP as a supplemental educational tool. These stakeholders benefit from KIP as a way to reinforce legal awareness outside the classroom.

3.3 Future Users

Future users of KIP may be members of juvenile diversion programs, community youth groups, and driving schools. These groups may use KIP as an educational resource to improve legal literacy in structured settings.

4. Glossary

CDC: Centers for Disease Control and Prevention

KIP: Knowledge Is Power

MFCD: Major Functional Component Diagram

UX: User Experience

5. References

Centers for Disease Control and Prevention. (2026). Transportation safety. Retrieved from

<https://www.cdc.gov/transportation-safety/index.html>

Legal Services Corporation. (2017). The justice gap: Measuring the unmet civil legal needs of low-income Americans. Retrieved from

<https://www.lsc.gov/sites/default/files/images/TheJusticeGap-FullReport.pdf>

National Highway Traffic Safety Administration. (2024). Teen driving: Talk to your teen driver about safe driving. Retrieved from

<https://www.nhtsa.gov/teen-driving/parents-talk-yourteen-driver-about-safe-driving>

Touro Law Jacob D. Fuchsberg Law Center. (2017). A lawyer's experience in K-12 law-related education: Lessons and opportunities. Retrieved from

<https://digitalcommons.tourolaw.edu/cgi/viewcontent.cgi?article=1023&context=jel>